

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- D1 cannot work Night shift
- D2 must work before D3 (Morning < Afternoon < Night)
- Only one doctor per shift
- D3 cannot work Morning
- All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, Right
- Cost of each move = 1
- Cannot pass through obstacles
- Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I P.Ramesh (192312443) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Ramesh

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

1)ANS;

Given

Doctors: D1, D2, D3

Shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work Night.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work Morning.
5. Every doctor must be assigned exactly one shift.

Step 1: Assign D1

Possible shifts:

- Morning
- Afternoon
- Night

D1 = Morning

Current Assignment:

D1 → Morning

D2 → ?

D3 → ?

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Try:

D2 = Afternoon

Current Assignment:

D1 → Morning

D2 → Afternoon

D3 → ?

Step 3: Assign D3

Remaining shift:

- Night

Check constraints:

- D3 cannot work Morning
- D2 before D3 (Afternoon < Night)
- One doctor per shift
- All doctors assigned

Therefore,

D3 = Night

2)ANS;

Problem Statement

A robot operates in a 5×5 grid and must move from the Start (S) position to the Goal (G) position.

Constraints

1. Movement allowed: Up, Down, Left, Right
2. Cost of each move = 1
3. Robot cannot move through obstacles (X)
4. Use Breadth-First Search (BFS) to find the shortest path.

Grid Representation

	C1	C2	C3	C4	C5
R1	S	.	.	X	.
R2	.	X	.	X	.
R3	.	X	.	.	.
R4	.	.	X	X	.
R5	.	.	.	.	G

- S = Start
- G = Goal
- X = Obstacle
- . = Free Path

Step 1: Initial State

- Queue = [S]
- Visited = {S}

Step 2: BFS Node Expansion;

Step	Expanded Node	Queue After Expansion
1	S(1,1)	[(1,2),(2,1)]
2	(1,2)	[(2,1),(1,3)]

Step	Expanded Node	Queue After Expansion
3	(2,1)	[(1,3),(3,1)]
4	(1,3)	[(3,1),(2,3)]
5	(3,1)	[(2,3),(4,1)]
6	(2,3)	[(4,1),(3,3)]
7	(4,1)	[(3,3),(5,1),(4,2)]
8	(3,3)	[(5,1),(4,2),(3,4)]
9	(5,1)	[(4,2),(3,4),(5,2)]
10	(4,2)	[(3,4),(5,2)]
11	(3,4)	[(5,2),(3,5)]
12	(5,2)	[(3,5),(5,3)]
13	(3,5)	[(5,3),(4,5)]
14	(5,3)	[(4,5),(5,4)]
15	(4,5)	[(5,4),(5,5)]
16	(5,4)	[(5,5)]
17	(5,5)	Goal Found

Queue Updates;

Queue = [S]

↓

[(1,2),(2,1)]

↓

[(2,1),(1,3)]

↓

[(1,3),(3,1)]

↓

[(3,1),(2,3)]

↓

...

↓

[(5,5)] → Goal Reached

Optimal Path;

S

↓

(2,1)

↓

(3,1)

↓

(4,1)

↓

(5,1)

→ (5,2)

→ (5,3)

→ (5,4)

→ G

Path Cost

Number of moves = 8

Cost per move = 1

Total Cost = 8

3)ANS;

Problem

Initial State

- Robot starts at S.

Goal State

- Reach the survivor at G.

State Space

- All accessible rooms (grid cells) in the building.
- Move Up
- Move Down
- Move Left
- Move Right

Constraints

- Cannot move through obstacles.
- Can observe only adjacent cells.
- Movement cost = 1.
- Entering a risky zone costs $1 + 2 = 3$.
- Robot must update its map dynamically (online search).

Search Strategy

The most suitable algorithm is A* (A-Star) Search with Manhattan Distance as the heuristic because:

- It finds the minimum-cost path.
- It avoids unnecessary exploration.
- It works efficiently with dynamic updates.

Cost Function

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ = Cost from Start to current node
- $h(n)$ = Manhattan Distance to Goal

Example Grid

```
S . X .  
. R . .  
. X . G  
. . . .
```

- S = Start
- G = Goal
- X = Obstacle
- R = Risky Zone

Step-by-Step Search

Step 1

Start at S

Open List = {S}

Cost = 0

Step 2

Expand neighboring cells.

Possible moves:

- Right
- Down

Choose the node with the lowest $f(n)$.

Step 3

Robot discovers a blocked path.

Update map.

Ignore blocked node.

Expand the next best node.

Step 4

Robot encounters a risky zone.

Entering Risk Zone:

Cost = $1 + 2 = 3$

If another safe path has a lower total cost, choose that path.

Step 5

Continue expanding nodes until Goal is reached.